Can AI Agents Answer Your Data Questions? A Benchmark for Data Agents

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Research Questions

- How well do current frontier LLMs perform on realistic, end-to-end data agent tasks that require multi-database integration, unstructured text processing, and domain knowledge?
- What are the primary failure modes of AI agents when querying enterprise-like data across heterogeneous database systems?
- What benchmark properties distinguish real-world enterprise data queries from existing text-to-SQL and table QA benchmarks?

Core Idea

Existing benchmarks for data agents only test isolated capabilities like SQL generation or table reasoning, but real enterprise data is messy - spread across multiple databases with inconsistent formats and critical information buried in unstructured text. This paper creates DAB, the first benchmark that tests end-to-end data agent workflows by systematically corrupting clean datasets to mirror enterprise messiness: distributing data across multiple database systems, reformatting join keys to be inconsistent, and embedding structured values in free text that agents must extract. The key insight is that agents fail not because they can't write SQL, but because they struggle with planning multi-step workflows and implementing robust text extraction - problems invisible to current benchmarks.

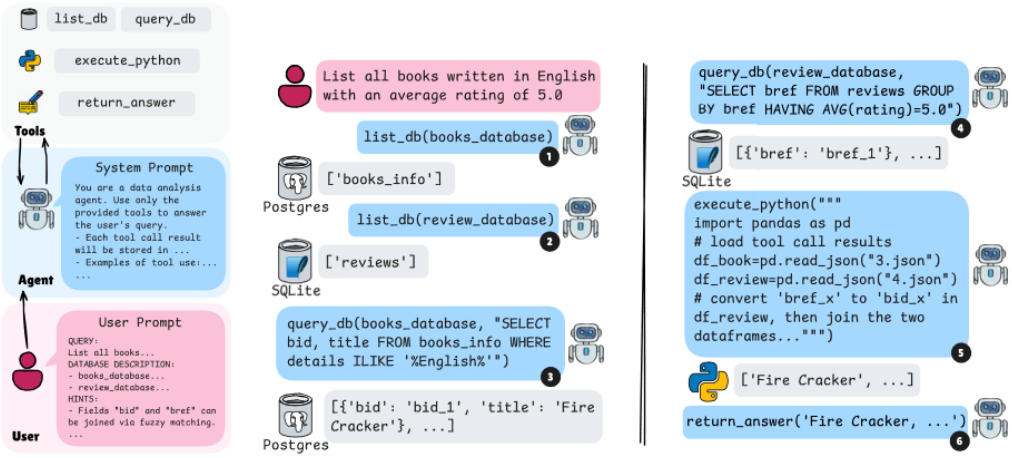
Key Formula

Pass@k accuracy metric

$$\text{pass@}k = 1 - \frac{\binom{n-c}{k}}{\binom{n}{k}}$$

Where n=50 total trials and c=correct trials. Estimates probability that at least one of k independent attempts succeeds. Used to measure agent accuracy across multiple attempts, distinguishing between queries that are solvable but unreliable (low pass@1, high pass@k) versus completely unsolvable (low pass@k for all k).

Key Figure



(a) DAB setup. (b) Example agent trace.

Figure 1: (a) In DAB, an agent solves a user task by interacting with database querying and Python execution tools within a ReAct-style loop. (b) In this example, the agent *operates over unstructured text* (i.e., extracting language from the `details` column in the 3rd tool call) and *integrates data across different databases* (PostgreSQL and SQLite) by reconciling the *ill-formatted join keys* (i.e., `bref` and `bid`, in the 5th tool call).

Figure 1: Shows the complete DAB agent architecture and concrete example of agent operating over unstructured text and integrating data across different databases by reconciling ill-formatted join keys.

Limitations

- Benchmark limited to 54 queries across 12 datasets, smaller than some existing benchmarks due to manual curation requirements
- Ground truth validation favors recall over precision - agents returning correct values alongside incorrect ones still pass validation
- Excludes open-ended analytical reasoning queries and live API integration to maintain deterministic evaluation
- All agents fail completely on patents dataset requiring complex date parsing from natural language, indicating fundamental extraction limitations